55. 12-Bit D/A Converter (DAC12)

55.1 Overview

This is the DAC_B version of the DAC12 peripheral module. DAC_B is referred to as DAC12 in this chapter.

The MCU provides a 12-bit D/A Converter (DAC12). Table 55.1 lists the DAC12 specifications, Figure 55.1 shows a block diagram, and Table 55.2 lists the I/O pins.

Table 55.1 DAC12 specifications

Parameter	Specifications
Resolution	12 bits
Output channels	2 channels
Module-stop function	Module-stop state can be set to reduce power consumption
Event link function (input)	The DA0 and DA1 conversion can be started on input of an event signal
Destination of D/A output control function	Controls whether the output to the external pin or to the internal modules (ACMPHS and ADC16H) is used
TrustZone Filter	Security and Privilege attribution can be set

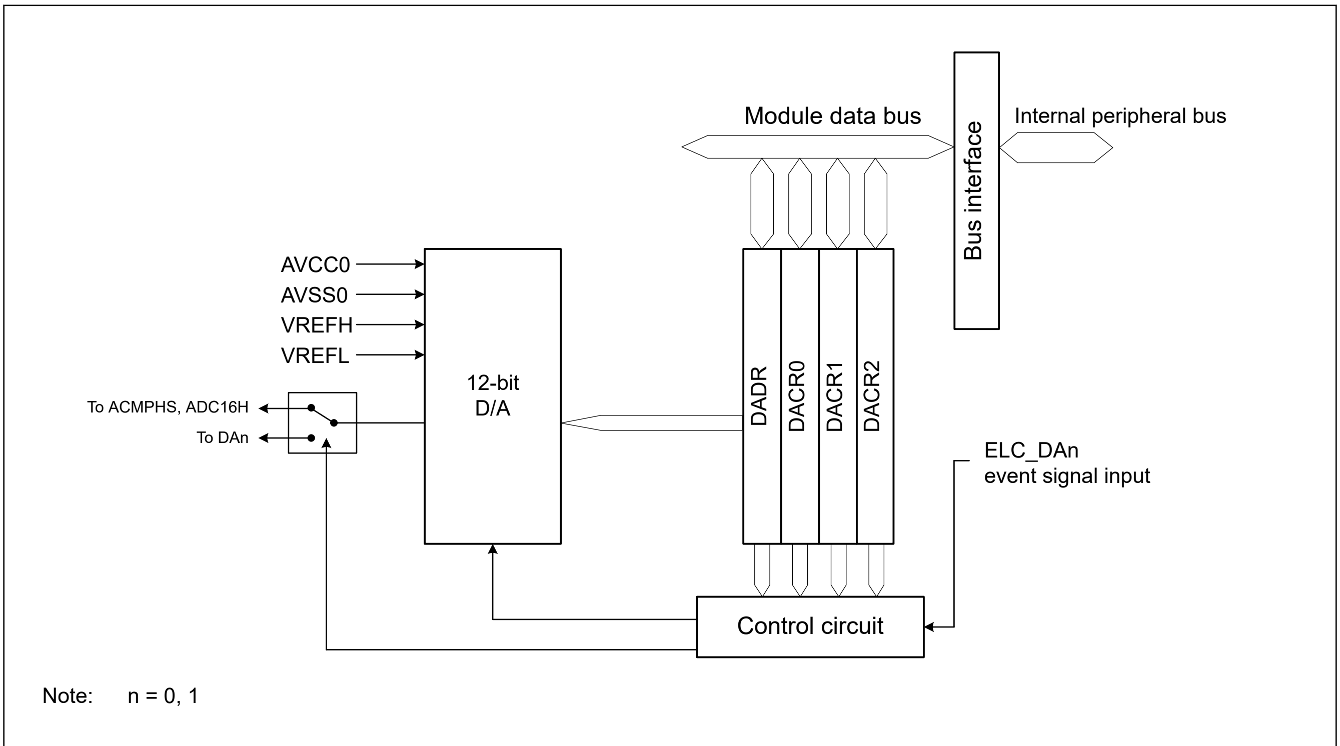

Figure 55.1 DAC12 block diagram

Table 55.2 lists the pin configuration of the DAC12.

Table 55.2 DAC12 I/O pins (1 of 2)

Pin name	I/O	Function
AVCC0	Input	● Analog power and analog reference top voltage supply pin for ADC16H and DAC12. ● Connect to VCC when these modules are not used.
AVSS0	Input	● Analog ground and analog reference ground supply pin for ADC16H and DAC12. ● Connect to VSS when these modules are not used.
VREFH	Input	Analog reference top voltage supply pin for the ADC16H (unit 1) and the DAC12
VREFL	Input	Analog reference ground pin for the ADC16H (unit 1) and the DAC12

Table 55.2 DAC12 I/O pins (2 of 2)

Pin name	I/O	Function
DA0	Output	Channel 0 output pin for the analog signals processed by the DAC12
DA1	Output	Channel 1 output pin for the analog signals processed by the DAC12

55.2 Register Descriptions

55.2.1 DADR : D/A Data Register

Base address: DACn_B = 0x4023_3000 + 0x0100 × n (n = 0, 1)
DACn_B_NS = 0x5023_3000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0000

Bit position:	15															0
Bit field:																
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Note: S-TYPE-3, P-TYPE-3

DADR register is 16-bit read/write registers that store data for D/A conversion. When an analog output is enabled, the values in DADR are converted and output to the analog output pins.

12-bit data can be formatted as left- or right-justified in the DACR1.DPSEL bit setting. In right-justified format (DACR1.DPSEL = 0), the lower 12 bits, [11:0], are valid. In left-justified format (DACR1.DPSEL = 1), the upper 12 bits, [15:4], are valid.

55.2.2 DACR0 : D/A Control 0 Register

Base address: DACn_B = 0x4023_3000 + 0x0100 × n (n = 0, 1)
DACn_B_NS = 0x5023_3000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	DAOUTDIS	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DAE*1	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DACEN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	DACEN	D/A Output Enable 0: Disable analog output of channel n (DAn) 1: Enable D/A conversion of channel n (DAn)	R/W
14:1	—	These bits are read as 0. The write value should be 0.	R/W
15	DAE*1	D/A Enable R/W 0: Control D/A conversion of channels 0 and 1 individually 1: Control D/A conversion of channels 0 and 1 collectively	R/W
30:16	—	These bits are read as 0. The write value should be 0.	R/W
31	DAOUTDIS	Analog Output Disables 0: The output to pin is enabled and the output to comparator is disabled 1: The output to pin is disabled and the output to comparator is enabled	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. This bit controls D/A conversion in combination with the DACR0.DACEN bit (n = 0, 1), which controls the output of the conversion results. For details, see Table 55.3.

Table 55.3 D/A conversion controls

DACR0.DAE of DAC120	DACR0.DAE of DAC121	DACR0.DACEN of DAC120	DACR0.DACEN of DAC121	Description
0	0	0	0	Disable D/A conversion and analog output pins (DA0, DA1)*1
0	0	1	0	• Enable D/A conversion of channel 0 and disable D/A conversion of channel 1 • Enable analog output of channel 0 (DA0) and disable analog output of channel 1 (DA1)*1
0	0	0	1	• Disable D/A conversion of channel 0 and enable D/A conversion of channel 1 • Disable analog output of channel 0 (DA0)*1 and enable analog output of channel 1 (DA1)
0	0	1	1	• Enable D/A conversion of channels 0 and 1 • Enable analog output of channels 0 and 1 (DA0, DA1)
1*2	x	x	x	• Enable D/A conversion of channels 0 and 1 • Collective enable analog output of channels 0 and 1 (DA0, DA1)
x	1*2	x	x	• Enable D/A conversion of channels 0 and 1 • Collective enable analog output of channels 0 and 1 (DA0, DA1)

Note: x: Don't care
Note 1. When analog output is disabled, the analog output signal is placed in the Hi-Z state.
Note 2. When security attribution of DAC120 and DAC121 does not match, setting 1 to DAE bit has no effect on D/A conversion and analog output pins.

DACEN bit (D/A Output Enable)

The DACEN bit controls D/A conversion and analog output in combination with the DAE bit. For details, see Table 55.4.

When both the DACEN bit and DAE bit are 0, D/A conversion of channel n (n = 0, 1) is not processed, and no conversion result is output.

DAE bit (D/A Enable R/W)

The DAE bit controls D/A conversion and analog output in combination with the DACEN bit. For details, see Table 55.4.

DAOUTDIS bit (Analog Output Disables)

The DAOUTDIS bit controls the analog output destination of the DAC12. Set 0 to output to the pin or 1 to output to the comparator. It is not possible to output to both the pin and the comparator at the same time.

The event link function can be used to set the DACEN bit to 1. The DACR0.DACEN bit of DAC120 is set to 1 when the event specified in the ELSR12 register of the ELC (ELC_DA0 event) occurs, and output of the D/A conversion results starts. The DACR0.DACEN bit of DAC121 is set to 1 when the event specified in the ELSR13 register of the ELC (ELC_DA1 event) occurs, and output of the D/A conversion results starts.

Table 55.4 D/A conversion and analog output control

DACR0.DAE	DACR0.DACEN	DACR0.DAOUTDIS	Channel n operation	Analog external output of channel n*1	Analog internal output of channel n*2
0	0	x	Stop	Hi-Z	Hi-Z
	1	0	Run	D/A output	Hi-Z
		1	Run	Hi-Z	D/A output
1	x	0	Run	D/A output	Hi-Z
		1	Run	Hi-Z	D/A output

Note: x : Don't care
Note 1. Output to pin
Note 2. Output to ACMPHS and ADC16H

55.2.3 DACR1 : D/A Control 1 Register

Base address: DACn_B = 0x4023_3000 + 0x0100 × n (n = 0, 1)
DACn_B_NS = 0x5023_3000 + 0x0100 × n (n = 0, 1)

Offset address: 0x0008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DPSEL
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
16	DPSEL	DADR Format Select R/W 0: Right-justified format 1: Left-justified format	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

DPSEL bit (DADR Format Select R/W)

The DPSEL bit controls the data format of the DADR register.

55.2.4 DACR2 : D/A Control 2 Register

Base address: DACn_B = 0x4023_3000 + 0x0100 × n (n = 0, 1)
DACn_B_NS = 0x5023_3000 + 0x0100 × n (n = 0, 1)

Offset address: 0x000C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	OFSSEL	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	OFSSEL	DAC Operating Voltage Mode Selection 0: Normal voltage mode (VREFH ≥ 2.7 V) 1: Low voltage mode (VREFH < 2.7 V)	R/W
31:9	—	These bits are read as 0. The write value should be 0.	R/W

OFSSEL bit (DAC Operating Voltage Mode Selection)

The OFSSEL bit controls the optimization of the DAC output voltage according to the analog reference voltage VREFH. When the VREFH voltage is less than 2.7 V, set the OFSSEL bit to 1.

55.3 Operation

The DAC12 performs D/A conversion when the DACR0.DAE bit is 1 or the DACR0.DACEN bit is 1. The output destination of the D/A conversion result can be switched by the DACR0.DAOUTDIS bit value.

When the DACR0.DAOUTDIS bit is 0, the D/A conversion result is output to the DAn pin in the DAC output mode. When the DACR0.DAOUTDIS bit is 1, the D/A conversion result is output to the comparator in the comparator output mode.

The output value (reference) is expressed by the following formula:

$$\frac{\text{Setting in DADR}}{4096} \times \text{VREFH}$$

55.3.1 Configuration Update Transfer the Coefficient Value

Before using the DAC12, the coefficient value should be transferred from extra MRAM. See section 60.10. Configuration Update Transfer.

55.3.2 DAC Output Mode

This following example shows D/A conversion on channel n (n = 0, 1) in DAC output mode. Figure 55.2 shows the timing of this operation.

1. Set the DACR0.DAOUTDIS bit to 1.
2. Set the DACR1.DPSEL bit and the DACR2.OFSSEL bit as required. Set the D/A conversion initial data in the DADR register.
   If the DACR2.OFSSEL bit is set to 0, then set 0x0E0 to the DADR register, otherwise set 0x1F8 to the DADR register as an initial value of starting up D/A conversion.
3. After t_{SU} elapses from the DAOUTDIS bit setting, set the DACR0.DACEN bit or the DACR0.DAE bit to 1.
4. After t_{DISOUT} elapses, set the DACR0.DAOUTDIS bit to 0. The D/A conversion result of the initial value is output to the DAn pin after the time of $t_{DSLPUP2}$.
5. Set the data for D/A conversion to the DADR register.
6. After t_{DCONV2} elapses, the D/A conversion result is output to the DAn pin.

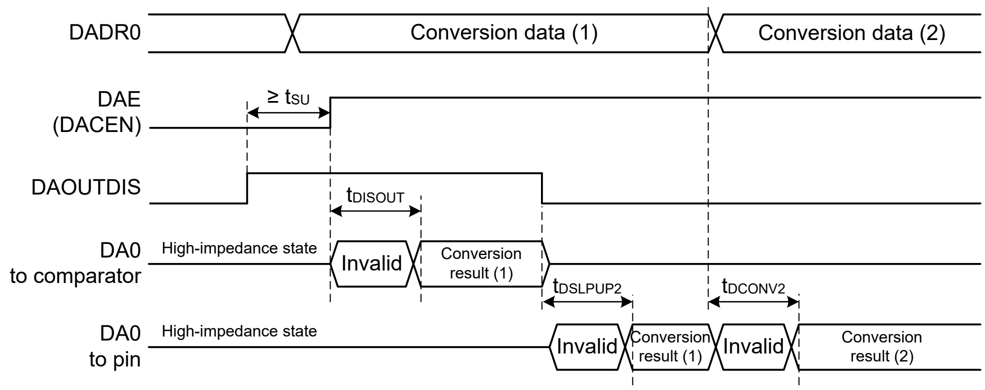

Figure 55.2 Example of DAC12 operation (DAC output mode)

55.3.3 Comparator Output Mode

This following example shows D/A conversion on channel n (n = 0, 1) in comparator output mode. Figure 55.3 shows the timing of this operation.

1. Set the DACR0.DAOUTDIS bit to 1.
2. Set the DACR1.DPSEL bit and the DACR2.OFSSEL bit as required. Set the D/A conversion initial data in the DADR register.
   If the DACR2.OFSSEL bit is set to 0, then set 0x0E0 to the DADR register, otherwise set 0x1F8 to the DADR register as an initial value of starting up D/A conversion.
3. After t_{SU} elapses from the DAOUTDIS bit setting, set the DACR0.DACEN bit or the DACR0.DAE bit to 1.

4. The D/A conversion result of the initial value is output to the comparator after the time of t_{DISOUT}.
5. Set the data for D/A conversion to the DADR register.
6. After t_{DCONV1} elapses, the D/A conversion result is output to the comparator.

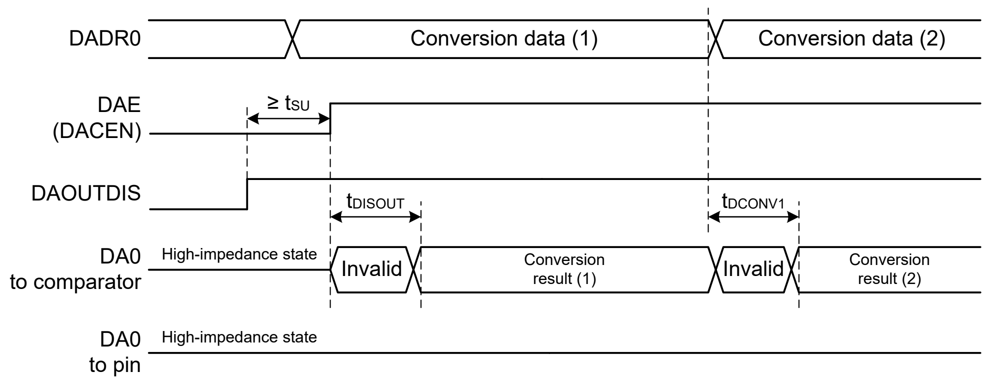

Figure 55.3 Example of DAC12 operation (comparator output mode)

55.4 Event Link Operation Setting Procedure

This section describes the procedures used in event link operation.

55.4.1 DA0 Event Link Operation Setting Procedure

To set up DA0 event link operation:

1. Set the DACR1.DPSEL bit to any value.
2. Set the DACR0.DACEN bit to 0.
3. Set the data for D/A conversion in the DADR.
4. Set the ELC_DA0 event signal to be linked to each peripheral module in the ELSR12 register.
5. Set the ELCR.ELCON bit to 1. This enables event link operation for all modules with the event link function selected.
6. Set the event output source module to activate the event link. After the event is output from the module, the DAC0DACR0.DACEN bit becomes 1, and D/A conversion starts on channel 0.
   The following time is required until the output value stabilizes.
   DAC output mode: 10.5µs
   Comparator output mode: 7µs
7. Set the ELSR12 register to 0x0000 to stop event link operation of DAC12 channel 0. All event link operation is stopped when the ELCR.ELCON bit is set to 0.

55.4.2 DA1 Event Link Operation Setting Procedure

To set up DA1 event link operation:

1. Set the DACR1.DPSEL bit to any value.
2. Set the DACR0.DACEN bit to 0.
3. Set the data for D/A conversion in the DADR.
4. Set the ELC_DA1 event signal to be linked to each peripheral module in the ELSR13 register.
5. Set the ELCR.ELCON bit to 1. This enables the event link operation for all modules with the event link function selected.
6. Set the event output source module to activate the event link. After the event is output from the module, the DACR0.DACEN bit becomes 1, and D/A conversion starts on channel 1.

The following time is required until the output value stabilizes.
DAC output mode: 10.5µs
Comparator output mode: 7µs

7. Set the ELSR13 register to 0x0000 to stop event link operation of DAC12 channel 1. All event link operation is stopped when the ELCR.ELCON bit is set to 0.

55.5 Analog Output Control by Security Attribution

Analog output to DA0/DA1 pin is controlled by the security attribution of Port and DAC12 as shown in Table 55.5.

Analog output to ACMPHS is not controlled by the security attribution.

Table 55.5 Analog output control by security attribution

Security Attribution			Analog Output	
DAC12 (PSARD)	**P014 (PmSAR (m = 0 to 9, A, B))**	**P015 (PmSAR (m = 0 to 9, A, B))**	**P014 pin output**	**P015 pin output**
0 (secure)	1 (non-secure)	1 (non-secure)	enable	enable
0 (secure)	1 (non-secure)	0 (secure)	enable	enable
0 (secure)	0 (secure)	1 (non-secure)	enable	enable
0 (secure)	0 (secure)	0 (secure)	enable	enable
1 (non-secure)	1 (non-secure)	1 (non-secure)	enable	enable
1 (non-secure)	1 (non-secure)	0 (secure)	enable	disable
1 (non-secure)	0 (secure)	1 (non-secure)	disable	enable
1 (non-secure)	0 (secure)	0 (secure)	disable	disable

55.6 Usage Notes on Event Link Operation

- When the event link function is used, set the DACR0.DAE bit to 0.
- When the event specified for the ELC_DA0 event signal is generated while writing to the DACR0.DACEN bit of DAC120 is performed, the write cycle is stopped, and the generated event takes precedence in setting the bit to 1.
- When the event specified for the ELC_DA1 event signal is generated while writing to the DACR0.DACEN bit of DAC121 is performed, the write cycle is stopped, and the generated event takes precedence in setting the bit to 1.

55.7 Usage Notes

55.7.1 Settings for the Module-Stop Function

DAC12 operation can be disabled or enabled using the Module Stop Control Register. The DAC12 is initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see section 11, Low Power Mode.

55.7.2 DAC12 Operation in the Module-Stop State

When the MCU enters the module-stop state with D/A conversion enabled, the D/A outputs are retained, and the analog power supply current is the same as during D/A conversion. If the analog power supply current must be reduced in the module-stop state, disable D/A conversion by setting the DACR0.DACEN and DACR0.DAE bits to 0.

55.7.3 DAC12 Operation in Software Standby Mode

When the MCU enters Software Standby mode with D/A conversion enabled, the D/A outputs are retained, and the analog power supply current is the same as during D/A conversion. If the analog power supply current must be reduced in Software Standby mode, disable D/A conversion by setting the DACR0.DACEN and DACR0.DAE bits to 0.

55.7.4 Constraint on Entering Deep Software Standby Mode

When the MCU enters Deep Software Standby mode with D/A conversion enabled, the D/A outputs are retained, and the analog power supply current is the same as during D/A conversion. If the analog power supply current must be reduced in Deep Software Standby mode, disable D/A conversion by setting the DACR0.DACEN and DACR0.DAE bits to 0.